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Deep learning algorithm for detection of aortic dissection on non-contrast-enhanced CT

Akinori Hata[1] · Masahiro Yanagawa[2] · Kazuki Yamagata[3] · Yuuki Suzuki[3] · Shoji Kido[3] · Atsushi Kawata[2] · Shuhei Doi[2] · Yuriko Yoshida[2] · Tomo Miyata[2] · Mitsuko Tsubamoto[1] · Noriko Kikuchi[2] · Noriyuki Tomiyama[2]

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Abstract

Objectives To develop a deep learning–based algorithm to detect aortic dissection (AD) and evaluate the diagnostic ability of the algorithm compared with those of radiologists.

Methods Included in the study were 170 patients (85 with AD and 85 without AD). An AD detection algorithm was developed using a convolutional neural network with Xception architecture. Of the patient data, 80% were used for training and validation and 20% were used for testing. Fivefold cross-validation was performed to evaluate the method. An average of 6688 non-contrast-enhanced CT images (slice thickness, 5 mm) were used for training. A radiologist reviewed both contrast-enhanced and non-contrast-enhanced images and identified the slices of AD. The identified slices were used as ground truth. Receiver operating characteristic curve and area under the curve (AUC) analysis was performed. Five radiologists independently evaluated the images. The accuracy, sensitivity, and specificity of the algorithm and those of the radiologists were compared.

Results The AUC of the developed algorithm was 0.940, and a cutoff value of 0.400 provided accuracy of 90.0%, sensitivity of 91.8%, and specificity of 88.2%. For the radiologists, median (range) accuracy, sensitivity, and specificity were 88.8 (83.5–94.1)%, 90.6 (83.5–94.1)%, and 94.1 (72.9–97.6)%, respectively. There was no significant difference in performance in terms of accuracy, sensitivity, or specificity between the algorithm and the average performance of the radiologists ($p > 0.05$).

Conclusions The developed algorithm showed comparable diagnostic performance to radiologists for detecting AD, which suggests the potential of the proposed method to support clinical practice by reducing missed ADs.

Key Points
- *A deep learning–based algorithm for detecting aortic dissection was developed using the non-contrast-enhanced CT images of 170 patients.*
- *The algorithm had an AUC of 0.940 for detecting aortic dissection.*
- *The accuracy, sensitivity, and specificity of the algorithm were comparable to those of radiologists.*

Keywords Aortic dissection · X-ray computed tomography · Artificial intelligence

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✉ Akinori Hata
a-hata@radiol.med.osaka-u.ac.jp

1 Department of Future Diagnostic Radiology, Graduate School of Medicine, Osaka University, 2-2 Yamadaoka, Suita, Osaka 565-0871, Japan

2 Department of Diagnostic and Interventional Radiology, Graduate School of Medicine, Osaka University, 2-2 Yamadaoka, Suita, Osaka 565-0871, Japan

3 Department of Artificial Intelligence Diagnostic Radiology, Graduate School of Medicine, Osaka University, 2-2 Yamadaoka, Suita, Osaka 565-0871, Japan

Abbreviations and acronyms

AD	Aortic dissection
AUC	Area under the curve
CAD	Computer-aided diagnosis
CI	Confidential interval
CT	Computed tomography
DICOM	Digital Imaging and Communications in Medicine
Grad-CAM	Gradient-weighted class activation mapping
HU	Hounsfield units
JPEG	Joint Photographic Experts Group
ROC	Receiver operating characteristic

Introduction

Aortic dissection (AD) is a common cardiovascular emergency that is often fatal. Its worldwide prevalence is estimated as 0.5–2.95 per 100,000 persons per year [1–3]. Contrast-enhanced computed tomography (CT) is the most commonly used diagnostic modality, yielding a sensitivity of 100% and a specificity of 98% [4, 5]. Although it is often difficult to detect AD on non-contrast-enhanced CT, these images can provide useful information such as displacement of intimal calcification and presence of high-attenuation false lumen [6].

The symptoms of AD are highly variable and may mimic more common conditions. Most patients with acute AD have sudden and severe chest, back, or abdominal pain [4], but some show only non-specific symptoms such as nausea or discomfort [7]. Up to 30% of patients who are ultimately diagnosed with AD may have had a different preliminary diagnosis [1–3]. If the symptoms are non-specific, AD can be missed on non-contrast-enhanced CT, although in some patients, it can be detectable by a careful retrospective review of the CT study. An automatic algorithm that can detect AD with high diagnostic ability would be particularly useful in clinical situations where reporting by a radiologist or consultation with a specialist is not always available.

Deep learning is a recently emerged technique that has application to various medical areas [8, 9]. Thoracic imaging is a good candidate for developing algorithms using deep learning technique. Numerous algorithms such as tuberculosis detection, nodule evaluation, and interstitial lung disease classification are being currently developed [10–13]. In this study, we developed a deep learning–based algorithm to detect AD on non-contrast-enhanced CT with slice thickness of 5 mm. We evaluated the diagnostic ability of the algorithm compared with those of radiologists.

Materials and methods

Patients

This retrospective study was approved by our institutional review board. Written informed consent was waived because of the retrospective study design. From the database in our institution, we retrieved the studies of 214 consecutive patients who underwent CT between January 2017 and April 2019 for the evaluation of AD. The principal investigator (A.H., with 7 years of radiology experience) reviewed the images of these patients, including previous CT studies, and identified those that met the following criteria: (a) CT obtained before aortic surgery and with no intra-aortic devices at the time of the study; (b) 5-mm slice thickness non-contrast-enhanced and contrast-enhanced images available, both obtained on the same day; and (c) AD detectable on contrast-enhanced CT for ≥ 15 mm in the *z*-axis direction. If several CT studies were available for the same patient, the most recent study was included. A final total of 85 studies were included as the positive cohort (AD studies). To obtain a cohort negative for AD, the principal investigator reviewed the database and identified CT studies that met the following criteria between September 2018 and April 2019: (a) the clinical indication for CT was pretreatment screening prior to cardiovascular surgery; (b) no intra-aortic devices at the time of the study; (c) 5-mm slice thickness non-contrast-enhanced and contrast-enhanced images were available, both obtained on the same day; and (d) contrast-enhanced CT revealed no AD or mural thrombus that was difficult to differentiate from AD. Eighty-five studies were included as the negative cohort (non-AD studies). A total of 170 studies conducted between November 2005 and April 2019 were included (Fig. 1). The patient demographics are summarized in Table 1. The patients were assigned to five cohort patterns to perform fivefold cross-validation (supplementary material). The cohort split (training/validation/test) was performed based on patients. Each of training, validation, and test sets did not have any slices from patients who were included in the other sets.

Image-based AD detection algorithm

Image selection

The principal investigator (A.H., with 7 years of radiology experience) reviewed both the contrast-enhanced and non-contrast images and identified the range of slices in which AD could be detected among the non-contrast images. The range of slices in which the aorta could be detected (from the top of the aortic arch to the bifurcation) was also identified. Figure 2 shows a summary of the image selection process. In the AD studies, images showing AD were regarded as positive images; in the non-AD studies, images that showed the aorta were regarded as negative images. Images in which the aorta was not present were not used, in both the AD and non-AD studies. In each of the training, validation, and test sets, the positive images were extracted and categorized as a positive set. Negative images were randomly selected so that the number of negative images equaled that of the positive images, and they were categorized as a negative set. Accordingly, the same numbers of positive and negative images were extracted in each set.

Development of the algorithm

Each of the images in Digital Imaging and Communications in Medicine (DICOM) format was converted to Joint Photographic Experts Group (JPEG) format with image preprocessing (supplementary material). The deep learning algorithm was developed using TensorFlow (version 1.14.0). The

Table 1 Patient demographics

	Total patients	AD patients	Non-AD patients
Number	*n* = 170	*n* = 85	*n* = 85
Age (years), mean ± SD	72.4 ± 11.0	69.2 ± 11.8	75.6 ± 9.0
Sex (female:male)	64:106	29:56	35:50
Background disease		AD (*n* = 85)	Valvular heart disease (*n* = 44)
			Aortic aneurism (*n* = 31)
			Coronary artery disease (*n* = 10)

AD aortic dissection

convolutional neural network used in this study was the Xception architecture, which is a variant of the Inception network [14]. We applied transfer learning using the weights learned on ImageNet dataset, thus enabling the network to understand basic shapes, such as edges, and their composition [15]. The first 108 layers of Xception architecture were frozen. We trained the algorithm using 2D images rather than 3D volume data because we used 5-mm-thick images that were not suitable for 3D-based analysis and because 2D-based training enabled us to simplify the image preprocessing. As mentioned above, fivefold cross-validation was performed, and five algorithm models were obtained.

Aorta detection algorithm

Using the abovementioned image-based AD detection algorithm, an image slice that included the aorta could be classified as positive or negative. It was assumed that this algorithm would not be able to correctly classify images that did not include the aorta. To achieve fully automatic evaluation by the algorithm, it was necessary to remove images that did not include the aorta. Therefore, we developed an additional aorta detection algorithm using the same five cohort patterns. In the development of this algorithm, images in which the aorta was seen were

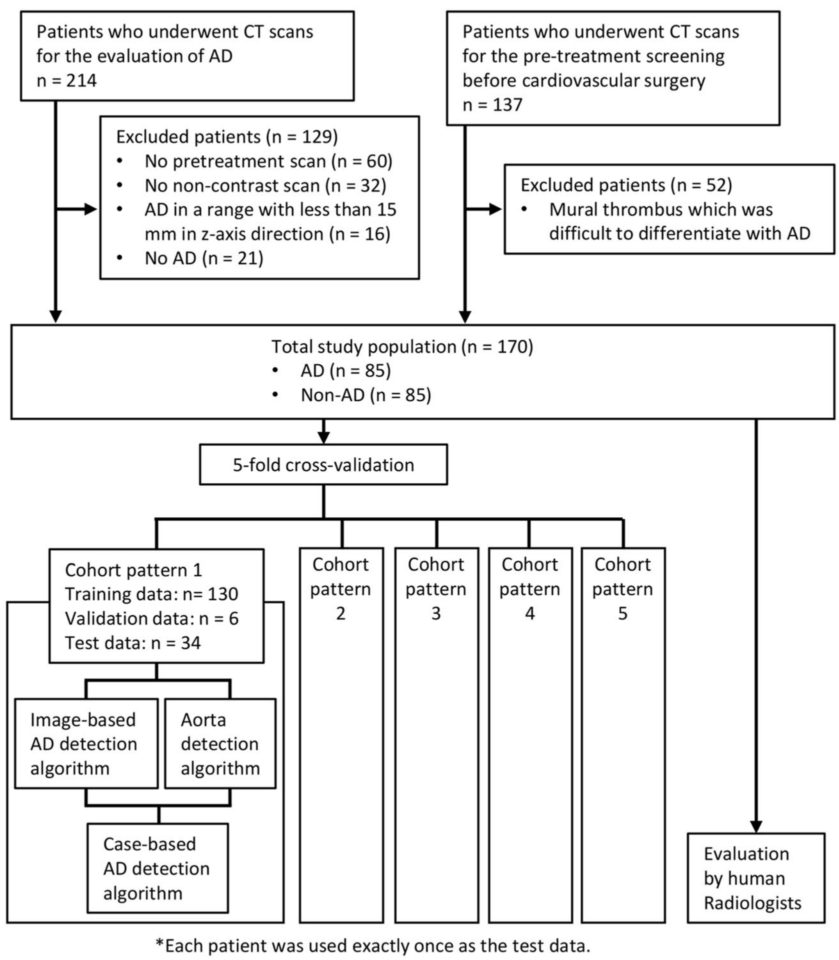

Fig. 1 Workflow of patient selection and evaluation. AD, aortic dissection; CT, computed tomography

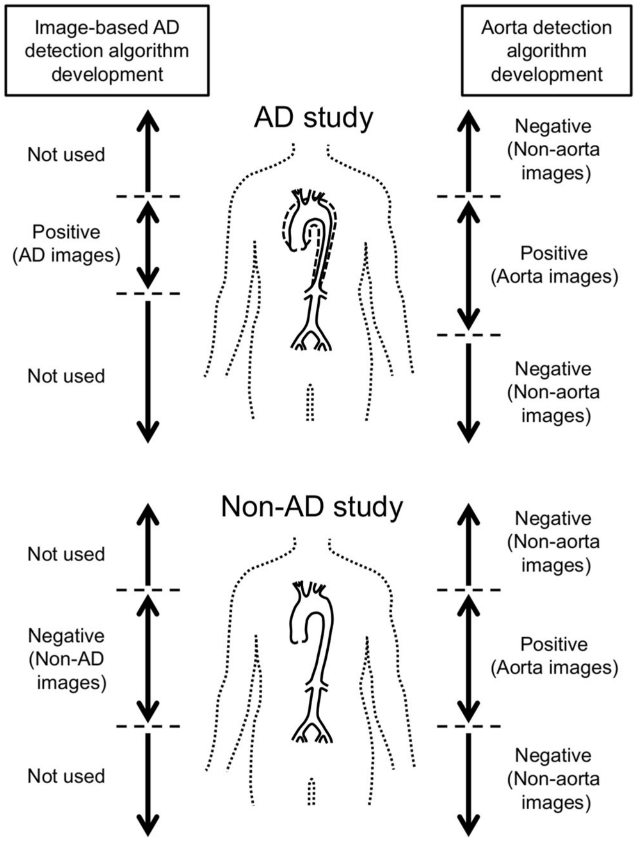

Fig. 2 Image selection in developing the image-based aortic dissection (AD) detection algorithm and the aorta detection algorithm

regarded as positive, and those in which the aorta was not seen were regarded as negative. The aorta detection algorithm was developed using the same architecture as previously, and fivefold cross-validation was performed.

Study-based AD detection algorithm

Using the abovementioned image-based AD detection and aorta detection algorithms, we then developed a study-based AD detection algorithm to classify the entire study as AD-positive or AD-negative. For each image, prediction values from 0 to 1 were obtained when the image was tested using each of the developed image-based AD detection algorithm and the aorta detection algorithm. A high prediction value indicates a high likelihood of AD, or a high possibility that the aorta is seen in the image, for the respective algorithms. The images of each study were tested by the model that had been trained using the cohort set that did not include the study. If the images of the study met the criteria, the study was diagnosed as AD; otherwise, it was diagnosed as non-AD ((a) a prediction value of the aorta detection algorithm > 0.5, (b) a prediction value of the image-based AD detection algorithm above a certain threshold (t), and (c) a certain number (X) of consecutive image slices in the study meeting both criteria (a and b)). The number of consecutive slices (X) was changed from 1 to 20 in increments of 1. The threshold (t) was changed from 0.001 to 1.000 in increments of 0.001, and the receiver operating characteristic (ROC) curve and the area under the curve (AUC) were obtained for each value of X.

Diagnostic ability of radiologists

The diagnostic abilities of five radiologists (A.K., S.D., Y.Y., T.M., and M.T., with 1 years, 3 years, 5 years, 6 years, and 24 years of radiology experience, respectively) were tested and compared with those of the developed algorithm. The radiologists independently reviewed the non-contrast DICOM images at the entire range of slices in each study. Window level and width were set as 30/400 but could be adjusted freely. If one or more slices were considered to show the existence of AD, the study was diagnosed as AD.

Verification of AD detection

In the deep learning–based algorithm, the weights are automatically optimized to classify the positive and negative sets. The trained algorithms sometimes perform classifications according to the factors which the human developers do not assumed. We considered that there might be selection bias in our cohort and the image-based AD detection algorithm might classify the images according to the factors other than the dissection in the aorta. To evaluate whether the area of the aorta in the image contributes to the decision of the developed image-based AD detection algorithm, this verification process was performed using a masked image set.

In each AD study, one image was randomly extracted from the images in which AD was seen. In each non-AD study, one image was randomly extracted from the images in which the aorta was seen. The extracted images were preprocessed as described above for algorithm development. A total of 170 JPEG images were obtained (original non-masked image set). To obtain a masked image set, the area of the aorta in each image was entirely masked by a black rectangle (Fig. 3). If both the ascending and descending aorta could be seen in the image, two black rectangles were placed. The original non-masked image set and the masked image set were tested by the image-based AD detection algorithm, and the prediction value for each image was obtained. The images were tested by the model trained using the cohort set that did not include the image.

Statistical analysis

For both the image-based AD detection algorithm and the aorta detection algorithm, the mean accuracy value of the five

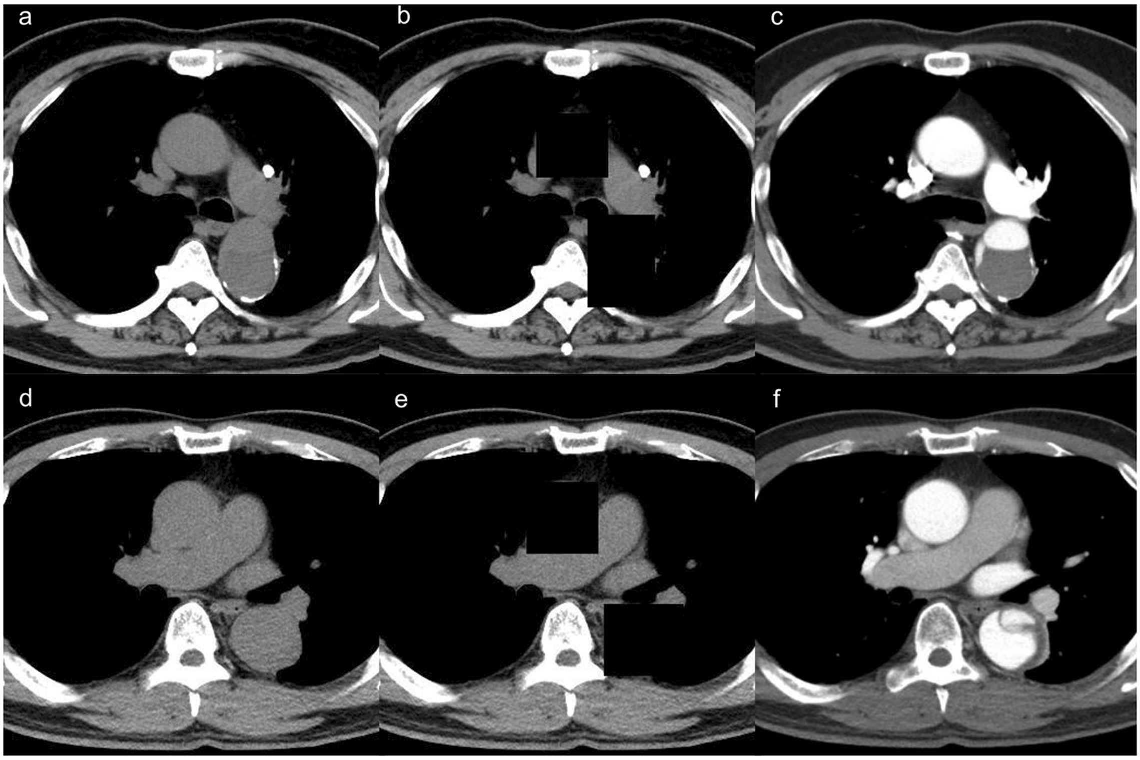

Fig. 3 Examples of original non-masked and masked images in 62-year-old (**a**–**c**) and 69-year-old males with aortic dissection (AD) (**d**–**f**). **a**, **d** Original non-masked images. **b**, **e** Masked images. **c**, **f** Contrast-enhanced images. The image-based AD detection algorithm correctly diagnosed the original non-masked images as AD. However, it diagnosed the masked images as non-AD

models for the fivefold cross-validation is presented as the total result. In the study-based AD detection algorithm, the best setting among the tested number of consecutive slices (X) for the optimal prediction value threshold (t) was determined by ROC curve analysis, and accuracy, sensitivity, and specificity were calculated.

To evaluate whether the algorithm performed better than the average performance of the radiologists, the differences in accuracy, sensitivity, and specificity were calculated between the algorithm and each radiologist, and these values (N = 5) were tested by one-sample t test. A p value of < 0.05 was considered to indicate statistically significant difference. In addition, differences in accuracy, sensitivity, and specificity between the algorithm and each radiologist were tested statistically using the McNemar test with Bonferroni correction. A p value of < 0.0083 (0.05/6) was considered to indicate statistically significant difference.

In verification of AD detection, ROC analysis was used to determine the AUC of each of the non-masked and masked sets. The two AUCs were compared by the DeLong method [16]. A p value of < 0.05 was considered to indicate statistically significant difference.

All statistical calculations were performed in Python and R (version 3.4.1; https://www.r-project.org/) environments.

Results

An average of 6688, 311, and 1750 images were used in the training, validation, and testing, respectively, of the image-based AD detection algorithm, and the processing time was 2714 s for training and 44 s (0.025 s/slice) for testing. An average of 17440, 806, and 4551 images were used in the training, validation, and testing, respectively, of the aorta detection algorithm, and the processing time was 5323 s for training and 105 s (0.023 sec/slice) for testing. In the training sets, the median (range) accuracies were 100.0 (100.0–100.0)% for the image-based AD detection algorithm and 100.0 (100.0–100.0)% for the aorta detection algorithm. In the test set, the median (range) accuracies were 86.8 (81.9–97.5)% for the image-based AD detection algorithm and 99.2 (99.2–99.4)% for the aorta detection algorithm.

The AUCs of the study-based AD detection algorithm are listed in Table 2 and shown in Fig. 4. The highest AUC value was 0.940 ($X = 9$). When the threshold of the prediction value (t) of the image-based AD detection algorithm was set to 0.400, accuracy, sensitivity, and specificity were 90.0 (95% confidential interval (CI) = 84.5–94.1)%, 91.8 (95% CI = 85.9–97.6)%, and 88.2 (95% CI = 81.4–95.1)%, respectively.

The results of the diagnostic ability of the radiologists and the algorithm are summarized in Table 3. The median (range) accuracy, sensitivity, and specificity of the five radiologists

Table 2 AUCs of the study-based aortic dissection (AD) detection algorithm

Number of consecutive slices (X)	AUC
1	0.825
2	0.891
3	0.901
4	0.908
5	0.927
6	0.928
7	0.933
8	0.931
9	0.940
10	0.927
11	0.929
12	0.924
13	0.924
14	0.921
15	0.919
16	0.933
17	0.933
18	0.934
19	0.923
20	0.923

AUC area under the curve

were 88.8 (83.5–94.1)%, 90.6 (83.5–94.1)%, and 94.1 (72.9–97.6)%, respectively. The average difference and the 95% CI for accuracy, sensitivity, and specificity between the algorithm and each radiologist were 0.6 (− 4.7 to 5.9)%, 2.6 (− 2.4 to 7.6)%, and − 1.4 (− 13.8 to 11.0)%, respectively, and there was no significant difference in accuracy (p = 0.761), sensitivity (p = 0.219), or specificity (p = 0.766). In the comparison between the algorithm and each radiologist, there was no significant difference in terms of accuracy or sensitivity ($p > 0.05$). The specificity of the algorithm was significantly higher than that of reader 4 (p = 0.005). It was lower than those of readers 1, 3, and 5, but there was no significant difference with Bonferroni correction (p = 0.166 for reader 1 vs. the algorithm, p = 0.021 for reader 3 vs. the algorithm, and p = 0.083 for reader 5 vs. the algorithm; $p < 0.01$ was considered statistically significant). There was no significant difference in specificity between the algorithm and reader 2 (p = 1.00). Representative images of patients with and without AD are shown in Fig. 5.

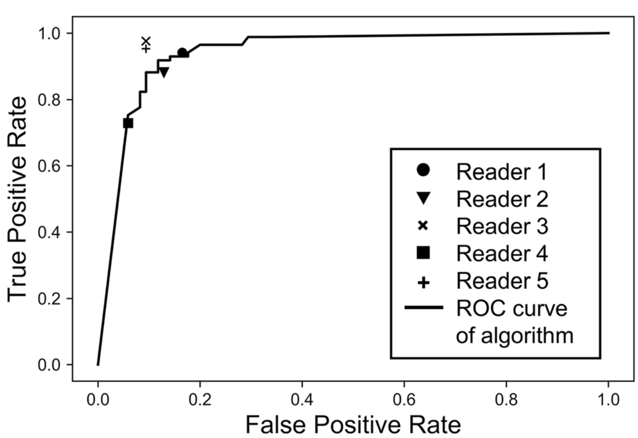

Fig. 4 ROC curve of the study-based aortic dissection (AD) detection algorithm, for studies that had 9 consecutive slices that were considered positive with both the image-based AD detection algorithm and the aorta detection algorithm. The area under the curve (AUC) was 0.940. The diagnostic abilities of the radiologists are indicated by markers (circle, triangle, x, rectangle, and cross). ROC, receiver operating characteristic

In the verification of AD detection, the AUC of the non-masked set was significantly higher than that of the masked set (non-masked set, 0.931; masked set, 0.761; $p < 0.001$; Fig. 3 and Supplementary Fig. A1), indicating that the masked area (aorta) significantly contributes to the decision of the algorithm. When the threshold prediction value was set to 0.500, the accuracy, sensitivity, and specificity were 85.3%, 78.8%, and 91.8% for the non-masked set and 68.8%, 42.3%, and 95.3% for the masked set, respectively.

Discussion

We developed a deep learning–based algorithm to detect AD using non-contrast-enhanced CT of 5 mm slice thickness. The developed algorithm had a sensitivity of 91.9% and a specificity of 85.8%, which were comparable to human diagnostic abilities. In the verification of AD detection, the AUC of the masked image set was significantly lower than that of the non-masked image set, indicating that the area of the aorta in the image significantly contributes to the decision of the algorithm. The developed algorithm may contribute to clinical practice by indicating to non-specialists the need to perform contrast-enhanced CT and by alerting radiologists to studies that have a high likelihood of AD. As the algorithm can indicate slice levels that are suspicious for AD, the clinicians can then perform a detailed review of the images.

Although there was no significant difference between the algorithm and each radiologist in terms of accuracy and sensitivity, the specificity of the algorithm showed a tendency to be slightly lower than that of three radiologists. However, this performance may be acceptable because we consider that the algorithm would be used for screening, in which sensitivity is more important. In the case that the algorithm indicated likely AD, further evaluation including contrast-enhanced CT would be performed.

In developing the algorithm, we used images of 5 mm slice thickness. We considered using thin-slice images, but these images are not always reconstructed in the clinical setting, particularly for patients with atypical symptoms. Algorithms developed using thin-slice images may achieve higher

Table 3 Diagnostic ability of the study-based aortic dissection (AD) detection algorithm and radiologists for the test set

	Radiology experience (years)	Accuracy, % (95% CI)	Sensitivity, % (95% CI)	Specificity, % (95% CI)
Reader 1	1	88.8 (83.1–93.1)	83.5 (75.6–91.4)	94.1 (81.4–95.1)
Reader 2	3	87.6 (81.7–92.2)	87.1 (79.9–94.2)	88.2 (81.4–95.1)
Reader 3	5	94.1 (89.4–97.1)	90.6 (84.4–96.8)	97.6 (94.4–100.0)
Reader 4	6	83.5 (77.1–88.8)	94.1 (89.1–99.1)	72.9 (63.5–82.4)
Reader 5	24	92.9 (88.0–96.3)	90.6 (84.4–96.8)	95.3 (90.8–99.8)
Algorithm		90.0 (84.5–94.1)	91.8 (85.9–97.6)	88.2 (81.4–95.1)

performance than the algorithm in this study; however, we chose a slice thickness of 5 mm to enable development of an algorithm that could be widely used. In addition, multiplanar reconstructions (MPRs) such as coronal and sagittal images or 3D volume image data may show different results. Further development using these data will require a larger study population because only a limited number of MPR or 3D dataset are obtained from each study.

The radiologists had high accuracies of 83.5–94.1% instead of using only non-contrast-enhanced CT. As the radiologists were asked specifically to detect AD, they evaluated the images in detail and may have be able to detect subtle findings

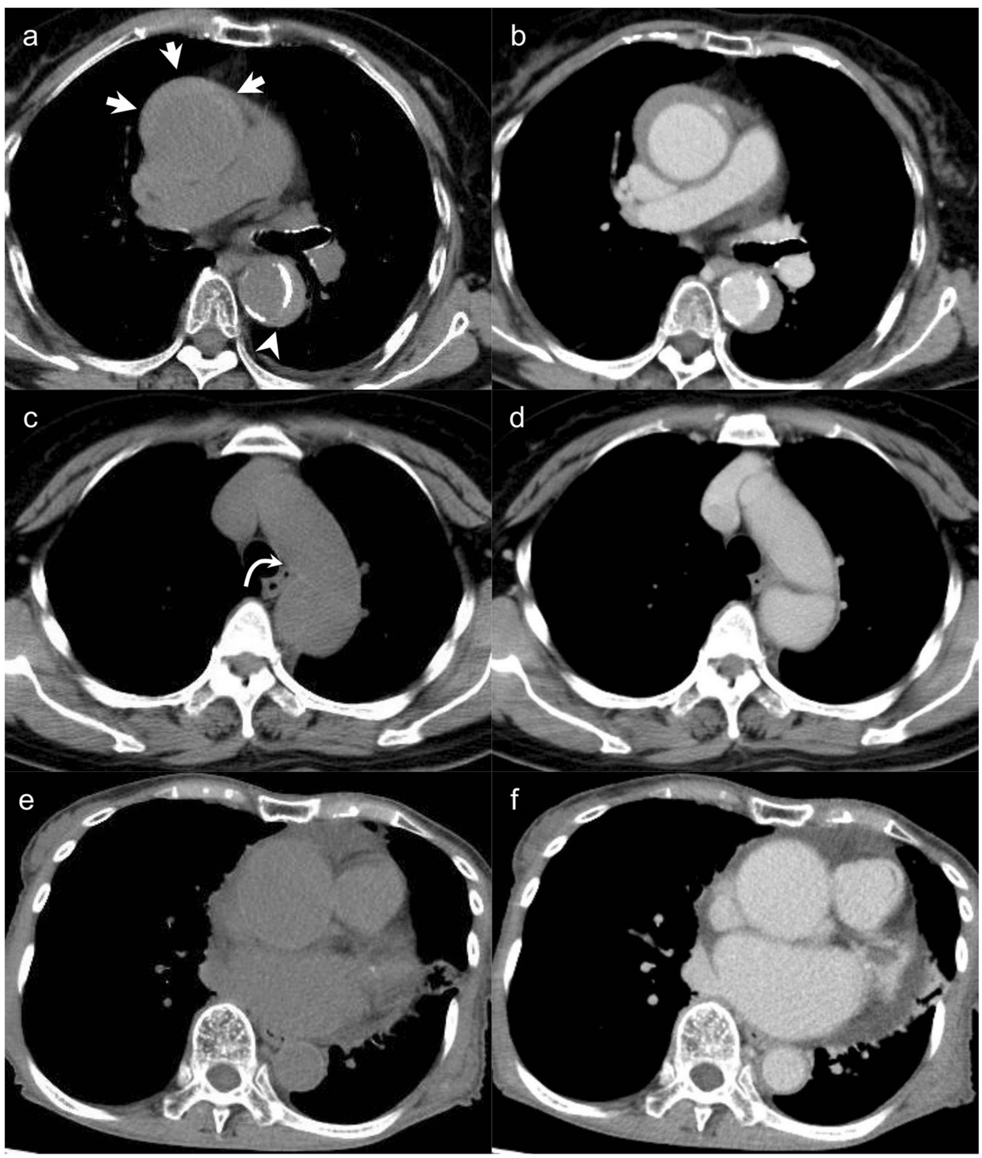

Fig. 5 An 80-year-old female with aortic dissection (AD) (**a**, **b**), a 69-year-old female with AD (**c**, **d**), and a 63-year-old male without AD (with aortic aneurism; **e**, **f**). **a**, **c**, **e** Non-contrast-enhanced images. **b**, **d**, **f** Contrast-enhanced images. Image **a** shows displacement of intimal calcification in the descending aorta (arrowhead) and also a high-attenuation area in the ascending aorta (arrows). The study-based AD detection algorithm and all radiologists diagnosed AD correctly. Image **c** shows a line of slightly high attenuation in the aortic arch (curved arrow) that suggests dissection. The algorithm diagnosed AD correctly, but only two of the five radiologists made a correct diagnosis. Image **e** shows an enlarged ascending aorta and pericardial effusion. The decision of the algorithm was negative for AD, and the decisions of the three radiologists were positive

such as displacement of tiny intimal calcification. This detailed assessment may have resulted in the high accuracies. In the clinical situation, the diagnostic ability of humans may be worse than our results, because clinicians may not consider a diagnosis of AD in patients with atypical symptoms and they may not have enough time to read the images in detail. In contrast, the automatic algorithm developed in this study provides stable performance even in asymptomatic patients, without the limitation of reading time.

In the verification of AD detection, the AUC of the masked image set was inferior to that of the non-masked image set but did not drop to 50%. As well as indicating a diagnosis of AD, the studies of these patients commonly reveal abnormal imaging findings in other regions. For example, pleural effusion occurs frequently in patients with acute AD [17]. AD can also cause organ infarction or ischemia due to arterial branch obstruction. Abdominal branch involvement has been reported in 27% of patients [18]. Conventional computer-aided diagnosis (CAD) systems comprise feature extraction, selection, and integration, which are complicated and laborious processes. In contrast, a deep learning algorithm learns the most predictive features of the desired outcome, without requiring detailed programming. The algorithm developed in the present study may therefore be using information about the features of regions other than the aorta to make predictions.

We introduced the concept of consecutive slices (X) in the study-based AD detection algorithm. The reason is that, in the case of $X = 1$, the diagnostic performance was not sufficient (the AUC was 0.825, and the accuracy, sensitivity, and specificity were 71.1%, 98.8%, and 43.5%, respectively, in the case of $t = 0.5$). Although $X = 9$ showed the best AUC, it may be controversial whether $X = 9$ is the best in the clinical setting. In the case of $X = 9$, localized ADs in 8 slices or less (4.0 cm in the craniocaudal direction) are always misdiagnosed. Less consecutive slices like $X = 5$ (AUC 0.927) may be appropriate in the clinical setting.

A major problem in developing deep learning algorithms is the so-called "black box phenomenon" [13]. As it is difficult to know how the algorithm has reached its conclusions, clinicians and radiologists might not be satisfied with the output. However, methods such as gradient-weighted class activation mapping (Grad-CAM) have been proposed to interpret the behaviors of deep learning algorithms [19]. Detailed interpretation of algorithm decision-making using such methods may enable new and clinically useful imaging findings to be captured.

Our study has several limitations. First, the study population was small. Overfitting is a major problem in machine learning and is more prominent when the number of training samples is small. Further validation of our algorithm using an external cohort is needed. Second, selection bias was inevitable because of the retrospective study design. Finally, we did not determine AD classification types such as acute/chronic and Stanford A/B. As clinical decision-making is highly influenced by type, an algorithm that could distinguish these types would make a greater contribution to clinical practice. To develop the algorithm, 3D-based approach with large size of cohort may be useful. In our 2D-based approach, it will be required to develop an algorithm to differentiate the ascending and descending aorta. However, contrast-enhanced CT will be performed for the diagnosis of AD and the type classification will be easier in contrast-enhanced images than non-contrast images.

In conclusion, the proposed algorithm could detect AD with a diagnostic performance comparable to that of radiologists, suggesting that it has potential to support clinicians.

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Compliance with ethical standards

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Statistics and biometry No complex statistical methods were necessary for this paper.

Informed consent Written informed consent was waived by the institutional review board.

Ethical approval Institutional review board approval was obtained.

Methodology
• retrospective
• diagnostic study
• performed at one institution

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